

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Articulation (Level 4)
Question Count: 5

Total Marks: 40

Code of Conduct

I **JAIVANT.S (192311326)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **JAIVANT.S**

Assessment Tasks

Industry Problem Solving Task

1. An e-commerce company receives clickstream, transaction, and review data from millions of users daily. Propose a big data analytics pipeline and explain how the 5Vs influence architecture design.
2. A healthcare organization collects structured patient records, semi-structured insurance claims, and unstructured medical images. Design an approach to manage and secure this data ecosystem.
3. A telecom company wants to detect churn patterns using high-volume customer interaction data. Identify the big data challenges and recommend suitable analytics and integration strategies.
4. A smart city project generates sensor streams from traffic, air quality, and surveillance systems. Explain how distributed file systems and data integration tools can support large-scale analysis.
5. A financial analytics firm must comply with strict privacy requirements while processing high-velocity transaction data. Design a secure big data solution and justify the controls chosen.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of	Good understanding	Basic understanding	Weak understanding

		real-world problem context			
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

ANSWERS

1. E-commerce Big Data Analytics Pipeline and 5Vs

An e-commerce company generates huge amounts of clickstream, transaction, and customer review data. A big data analytics pipeline can be designed as follows:

Data Sources:

Data is collected from websites, mobile applications, transaction systems, customer reviews, and social media.

Data Ingestion:

Tools such as Apache Kafka can collect real-time clickstream and transaction data. Batch data can also be uploaded periodically.

Data Storage:

A distributed storage system such as Hadoop HDFS or cloud object storage can store large volumes of structured and unstructured data.

Data Processing:

Apache Spark can process the data for cleaning, transformation, aggregation, and analysis.

Analytics:

Machine learning and analytics can be used for recommendation systems, customer segmentation, fraud detection, and sales prediction.

Visualization:

Tools such as Power BI or Tableau can display dashboards and business insights.

Influence of 5Vs:

- **Volume:** Millions of users generate huge amounts of data, requiring distributed storage.
- **Velocity:** Clickstreams and transactions arrive continuously, requiring real-time processing.
- **Variety:** Data includes structured transactions, semi-structured logs, and unstructured reviews.
- **Veracity:** Incorrect reviews, duplicate records, and incomplete data require data cleaning.
- **Value:** The processed data should provide useful insights such as personalized recommendations and customer behavior.

Thus, a distributed and scalable architecture is required to handle all five Vs efficiently.

2. Healthcare Big Data Management and Security

A healthcare organization handles different types of sensitive data, including patient records, insurance claims, and medical images. A suitable architecture should support different data formats while maintaining strong security.

Data Storage:

- Structured patient records can be stored in relational databases.
- Semi-structured insurance claims can be stored using NoSQL databases.
- Medical images can be stored in distributed object storage or cloud storage.

Data Integration:

ETL/ELT tools can collect data from hospitals, insurance systems, laboratories, and imaging systems and integrate it into a centralized data lake or warehouse.

Data Processing:

Apache Spark can process large datasets for medical research, reporting, and predictive analytics.

Data Governance: Maintain proper data ownership, metadata, data-quality rules, and retention policies for healthcare information.

Disaster Recovery: Replicate critical patient data across multiple locations so that it can be recovered during hardware failures or other disasters.

Security Measures:

- **Encryption:** Encrypt data both during transmission and storage.
- **Access Control:** Use role-based access control so employees access only required information.
- **Authentication:** Use strong authentication and multi-factor authentication.
- **Data Masking:** Hide sensitive patient information during analysis.
- **Auditing:** Maintain logs of data access and modifications.
- **Backup:** Maintain secure backups for disaster recovery.

This approach provides scalable data management while protecting sensitive patient information from unauthorized access.

3. Telecom Churn Detection Using Big Data

A telecom company generates large amounts of customer interaction data from calls, messages, internet usage, complaints, recharge history, and customer support.

Major Big Data Challenges:

- **High Volume:** Millions of customers generate huge datasets.
- **High Velocity:** Call and network events are generated continuously.
- **Variety:** Data comes from billing systems, call records, applications, and customer feedback.
- **Data Quality:** Missing, duplicate, or incorrect customer information can affect predictions.
- **Scalability:** The system must handle increasing numbers of customers.

Analytics Strategy:

Machine learning algorithms can identify customers who are likely to leave the service. Historical customer data can be used to train classification models such as decision trees, random forests, or logistic regression.

Important features can include:

- Call and internet usage
- Recharge frequency
- Customer complaints
- Service interruptions
- Subscription type
- Billing history

Integration Strategy:

Kafka can be used for real-time data ingestion, while Spark can process large datasets. Data from CRM, billing, network, and customer-support systems can be integrated into a central data lake.

Real-Time Churn Prediction: Streaming analytics can identify customers showing sudden changes in usage or complaints and generate immediate churn alerts.

Personalized Retention: Based on churn predictions, the company can provide personalized plans.

The final system can generate churn scores and help the company provide special offers or improved services to customers at high risk of leaving.

4. Distributed File Systems and Data Integration in Smart Cities

A smart city generates continuous data from traffic sensors, air-quality sensors, surveillance systems, public transportation, and other IoT devices.

Distributed File System:

Hadoop Distributed File System (HDFS) can store massive amounts of sensor and surveillance data across multiple machines. Data is divided into blocks and distributed across nodes. Replication provides fault tolerance if one node fails.

Data Ingestion:

Tools such as Apache Kafka can collect real-time streams from thousands of sensors.

Data Integration:

Data integration tools combine information from different sources such as traffic systems, environmental sensors, and surveillance systems. ETL processes can clean and transform the data before analysis.

Data Processing:

Apache Spark can perform large-scale batch and real-time processing.

Fault Tolerance: HDFS replication ensures that sensor data remains available even if individual storage nodes fail.

Analysis:

The processed data can be used to:

- Detect traffic congestion
- Monitor air pollution
- Identify unusual events
- Optimize public transportation
- Support emergency response

Benefits:

Distributed storage provides scalability and fault tolerance, while integration tools provide a unified view of data from different city systems.

Therefore, HDFS, Kafka, Spark, and data integration technologies can form an effective large-scale smart city analytics platform.

5. Secure Big Data Solution for Financial Analytics

A financial analytics company processes high-velocity transaction data while following strict privacy requirements. The architecture must provide security, scalability, and real-time processing.

Data Ingestion:

Apache Kafka can collect transactions continuously from banking and financial systems.

Storage:

Encrypted distributed storage or a secure cloud data lake can store transaction data. Sensitive information should be protected using encryption.

Processing:

Apache Spark can process transactions in real time and identify suspicious activities or fraud patterns.

Security Controls:

- **Encryption at Rest:** Protect stored financial data.
- **Encryption in Transit:** Secure data while moving between systems.
- **Role-Based Access Control:** Give users access only to the data required for their jobs.
- **Multi-Factor Authentication:** Prevent unauthorized account access.
- **Data Masking/Tokenization:** Hide sensitive account and customer information.
- **Audit Logs:** Record who accessed or modified data.
- **Data Loss Prevention:** Prevent sensitive information from being improperly copied or shared.
- **Backup and Disaster Recovery:** Protect against data loss and system failures.

These controls ensure confidentiality, integrity, and availability of financial data while allowing high-speed analytics. The solution can also support compliance with applicable financial data-protection and privacy regulations.